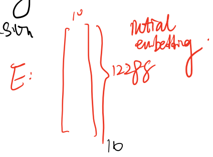
In 3b1b. $\underbrace{\overset{12288\times10}{(W_V E)}}_{12288\times12288}\ \text{soft}\left(\frac{\overset{128\times12288\ \ 12288\times10\ T}{(W_K E)(W_Q E)}}{\underset{10\times10}{\sqrt{d_k}}}\right) \to 12288\times10$

$T :=$ # of tokens: $T=10$

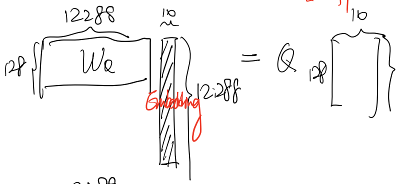
$H :=$ # of heads: 96

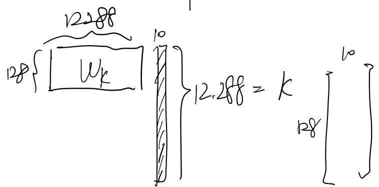
$d_{head} =$ dimension of / head width: 128
bottle neck.

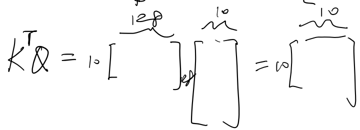
$d_{model} = H d_h := 12{,}288$
embedding
dimension


E:
10
12288
initial embedding

3 blue 1 brown


12288
10
128
W_Q
Embedding
12288
= Q
10
128

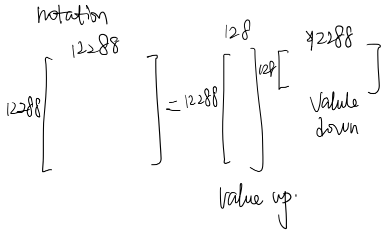

12288
10
128
W_K
12288 = K
10
128

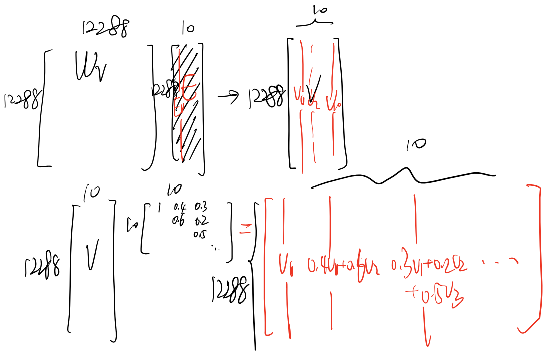

K^TQ = 10
128
10
128
= 10
10

$A = \text{softmax}\left(\frac{K^TQ}{\sqrt{head}}\right) := 10\times10$

$W_o W_v X$:

3b1b
notation


12288
12288
= 12288
128
128
12288
Value
down
value up.


12288
W_v
12288
10
12288
E
→ 12288
10
V_1
V_2
V_{10}
10
V
12288
10
10
1
0.4
0.3
0.6
0.2
0.5
=
12288
10
V_1
$0.4V_1+0.6V_2$
$0.3V_1+0.2V_2+0.5V_3$
…

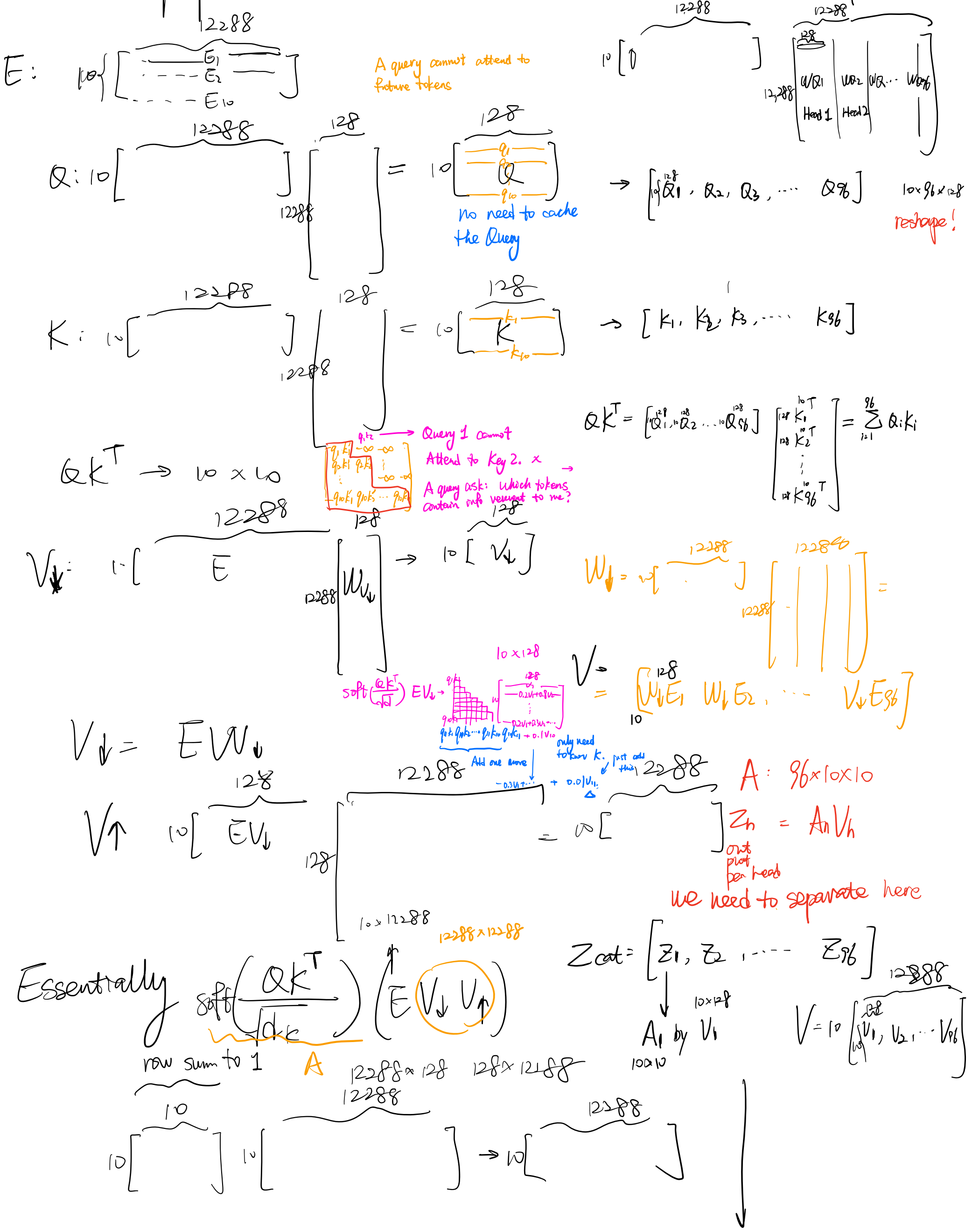
essentially $V_{12288\times12288}E_{12288\times10}\cdot A_{10\times10} \to 12288\times10$
↓
added to E

2^{nd} $AB = \begin{bmatrix} Ab_1 & Ab_2 & \cdots & Ab_n \end{bmatrix}$

In paper

formula $\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)$

In Reality


12288
E:
10
E_1
E_2
E_{10}
Q: 10
12288
128
12288
= 10
128
q_1
q_2
Q
q_{10}
K: 10
12288
128
12288
= 10
128
k_1
K
k_{10}
$QK^T \to 10\times10$
q_1k_1
$-\infty$
q_2k_1
q_2k_2
$q_{10}k_1$
$q_{10}k_2$
$q_{10}k_{10}$
q_1k_2
$V_\downarrow$: 10
12288
E
128
12288
$W_{V\downarrow}$
→ 10 [$V_\downarrow$]
$V_\downarrow = EW_\downarrow$
$V_\uparrow$ 10 [$EV_\downarrow$]
128
10×12288
12288
= 10
12288
10×128
$\text{soft}\left(\frac{QK^T}{\sqrt{d}}\right)EV_\downarrow \to$
q_1k_1
$q_{10}k_1$
$0.2V_1+0.8V_2$
$0.2V_1+0.3V_2\ldots$
$q_1k_1\ q_1k_2\ldots q_1k_{10}\ q_1k_1 + 0.1V_{10}$
Add one more
$0.3V_1\ldots$
$+\ 0.01V_{11}$
only need to know k.
just add this
12288
10
12288
W_{Q1}
W_{Q2}
$W_{Q}\ldots$
W_{Q96}
Head 1
Head 2
12288
$\to [Q_1, Q_2, Q_3, \ldots Q_{96}]$
10×96×128
reshape!
$\to [K_1, K_2, K_3, \ldots K_{96}]$
$W_\downarrow$ = 10
12288
12288
12288
=
V = [$W_\downarrow E_1$ $W_\downarrow E_2$, … $V_\downarrow E_{96}$]
128
10
12288×12288
Essentially
row sum to 1
A
12288×128
128×12288
10
10
10
12288
→ 10
12288
$Z_{cat} = [Z_1, Z_2, \ldots Z_{96}]$
12288
A_1 by V_1
10×10
10×128
$V = 10\,[V_1, V_2, \ldots V_{96}]$
128
10

A query cannot attend to future tokens

no need to cache the Query

Query 1 cannot Attend to Key 2. ×

A query ask: which tokens contain info relevant to me?

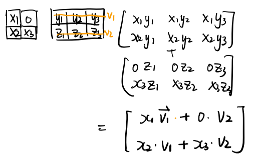
$QK^T = [Q_1, Q_2 \ldots Q_{96}]\begin{bmatrix} K_1^T \\ K_2^T \\ \vdots \\ K_{96}^T \end{bmatrix} = \sum_{i=1}^{96} Q_iK_i$

Essentially $\underbrace{\text{soft}\left(\frac{QK^T}{\sqrt{d_k}}\right)}_{A}\left(E\,V_\downarrow V_\uparrow\right)$

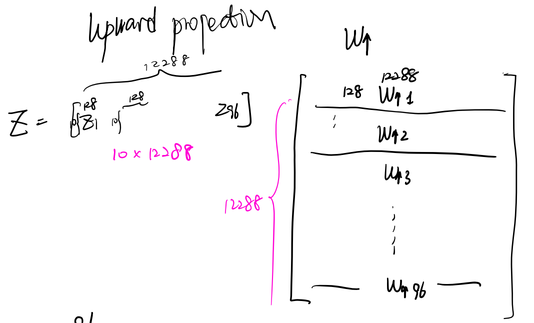
A: 96×10×10

$Z_h = A_hV_h$
output per head

we need to separate here


x_1 0
x_2 x_3
y_1 y_2 y_3
z_1 z_2 z_3
V_1
V_2
x_1y_1 x_1y_2 x_1y_3
x_2y_1 x_2y_2 x_2y_3
+
$0z_1$ $0z_2$ $0z_3$
x_3z_1 x_3z_2 x_3z_3
$= \begin{bmatrix} x_1\vec{V_1} + 0\cdot V_2 \\ x_2\cdot V_1 + x_3\cdot V_2 \end{bmatrix}$

Upward projection


12288
Z = [Z_1 Z_{96}]
128
10
10 × 12288
$W_\uparrow$
12288
128
$W_{\uparrow 1}$
$W_{\uparrow 2}$
$W_{\uparrow 3}$
$W_{\uparrow 96}$
12288

$= \sum_{i=1}^{96} Z_i W_{\uparrow i}$ (no longer needed to keep 96 heads anymore)

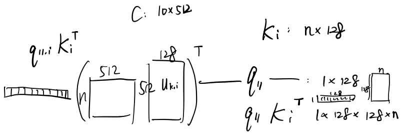
→

Multi Head MHA

per token 96×128

$2n_hd_hl$ =24576 2×12288

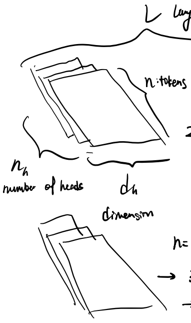
4MB


$q_{1,i}K_i^T$
C: 10×512
512
n
512
U_{Ki}
128
T
q_1
K_i: n×128
1×128
n
128
128
$q_1K_i^T$
1×128×128×n

1st regular dot product

3rd outer $\sum\left[\,\vdots\,\Big|\,\overline{\;x\;}\,\right]$ product sum

Keys Cache



Values Cache

Storage

$2\cdot n\cdot d_h\cdot n_h\cdot L$

$d_h=128$

$h_n=128$

$L=61$

$n=100000$ 100k token

→ 3.998×10^6 Bytes/token = 4MB/token

→ 400GB for each new token

Multi-Query Attention MQA

Share a Single K & V for all

$2d_h\cdot L$ 31 KB Less Specilization

$2\times128=256$

per token

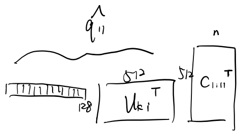
stores $\overbrace{\boxed{\quad k.\quad}}^{}_{128}$ $\boxed{\qquad}_{128}$

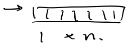
Group-Query GQA

groups share same KV.

multiple groups

$2n_g\cdot d_h\cdot L$ 500 KB





Multi-Head Latent Attention (MLA)

cache

512 per token

$d_L := 512$

latent dimension

$\underset{10\times12288}{E} \times \underset{12288\times512}{D_{KV}} \rightarrow \underset{10\times512}{C}$ Computed form of token

$\underset{10\times128}{\underbrace{k_i}} = \underset{10\times512}{C}\;\overset{\text{key features}}{\underset{512\times128}{U_{ki}}}$ $\qquad \overset{\text{value features}}{V_i = \underset{10\times512}{C}\cdot\underset{512\times128}{U_{v,i}}}$ (with V_i 10x128)

previous 12288×128

latent key features

$\underset{10\times128}{Q_i} = \underset{10\times12288}{\underbrace{E}}\;\underset{12288\times128}{\underbrace{W_{Qi}}}$ → Stays the same

$k_i^T = (CU_{ki})^T = \overset{512\times128}{U_{ki}^T}\,C^T$

$\hat{Q}_i$: 10×512 → Query projected to latent space

$Q_ik_i^T = \underset{10\times12288}{E}\;\underbrace{\underset{12288\times128}{W_{Qi}}\;\underset{128\times512}{U_{ki}^T}}\;\underset{512\times10}{C^T}$ → Attention Grid

Defined as B_i

12288×128 · 128×512 → 12288×512

Therefore $Q_ik_i^T := \underset{10\times512}{\underbrace{\hat{Q}_i}}\,C^T$

Attention pattern computed by multiplying the cached latent key value matrix

previously

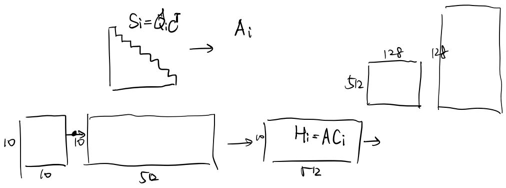
$k_i = \underset{10\times12288}{X}\cdot\underset{12288\times128}{W_{ki}} \rightarrow 10\times128$

$V_i = X\,W_{Vi}$ 10×12288×12288×128

$q_{10}k_1\;q_{10}k_2\cdots q_{10}k_{10}$ newly added

$\mathrm{soft}\left(\underset{10\times10}{\frac{QK^T}{\sqrt{d}}}\right)\underset{10\times128}{V}$

calculate new q_{10}



$Y_i = (A_iC)\,T_i$

$= \underset{10\times10}{(A_i}\,\underset{10\times512}{C)}\;\overset{T_i}{\underset{512\times128}{U_{vi}}\;\underset{128\times12288}{O_i}}$

A_iV_i

Average the latent vector first, then convert that average into value

$Z_{11,i} = a_1V_{1,i} + a_2V_{2,i} + \cdots + a_{11}V_{11,i}$

$V_{s,i} = c_sU_{vi}$ (s^{th} token)

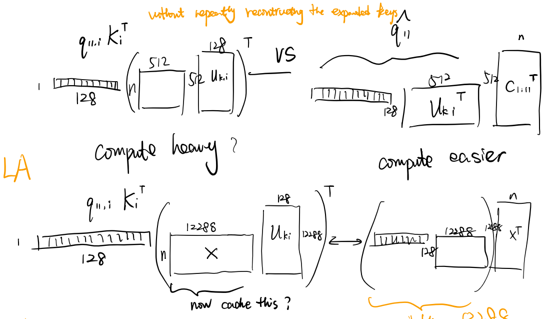
$= (a_1c_1 + a_2c_2 + \cdots + a_{11}c_{11})\,U_{v,i}$

don't need to construct CU_{vi} any more.

Instead use c caches.

let us retain the small latent cache without repeatly reconstructing the expanded keys

MLA



halves the cache size.

replace key/input cache by X:

however history-scoring dot product width: 12288

	cached number/token	width of history scoring dot product
MHA	2×96×128 = 24576	128
my proposal cache X without latent	12288	12288
MLA:	512	512

my proposal: $D_{KV} = I$: 12288

$C = E$

$U_{ki} = W_{ki}$ $\quad U_{vi} = W_{vi}$

Core: rearrange Query Computation

to absorb the key weights U_{ki} $\quad (W_Q\,W_{UK}^T)\,C^T$

rearrang final output computation to absorb the the value weights $U_{v\cdot i}$

why is it still called KV matrix?

$\hat{Q}_iC^T = (Q_iU_{ki}^T)\,C^T = Q\,((CK_{ui}))^T = QK_i^T$

$(A_iC)T_i = A(CU_{vi})O = A_i\,V_i\,O_i$ $\quad$ C still supplies historical info as values

C : representation of token

$\hat{q}_{11,i}$ current query transformed to score those representative

T_i Fixed learning weights for converting latent input into output.